DATASET INTELLIGENCE & OPERATIONAL ANALYSIS

Domain: Universal Dataset Intelligence & Distribution Analysis (0.0%) | Profiling Conf: 89.6% | Records: 300 | Hash: 0aa8ee84

Filter: ALL_RECORDS(UNFILTERED) | Audit: VERIFIED_DETERMINISTIC_PARITY | Engine: v8-javascript

KEY PERFORMANCE INDICATORS (DETERMINISTIC METRIC AUDIT)

TOTAL RECORDS VOLUME

300

`COUNTROWS`

Omitted: 0 rows

UNIQUE ENTITIES (CAMPAIGN_ID)

300

`COUNTDISTINCT(Campaign_ID)`

Omitted: 0 rows

TOTAL AD_SPEND_USD

1,176,908.40

`SUM(Ad_Spend_USD)`

Omitted: 0 rows

MEAN AD_SPEND_USD

3,923.03

`AVERAGE(Ad_Spend_USD)`

Omitted: 0 rows

MEDIAN AD_SPEND_USD

3,073.80

`MEDIAN(Ad_Spend_USD)`

Omitted: 0 rows

TOTAL COST_PER_ACQUISITION

7,498.74

`SUM(Cost_Per_Acquisition)`

Omitted: 0 rows

MEAN COST_PER_ACQUISITION

25.00

`AVERAGE(Cost_Per_Acquisition)`

Omitted: 0 rows

MEDIAN COST_PER_ACQUISITION

25

`MEDIAN(Cost_Per_Acquisition)`

Omitted: 0 rows

EXECUTIVE MANAGEMENT DECISION BRIEF

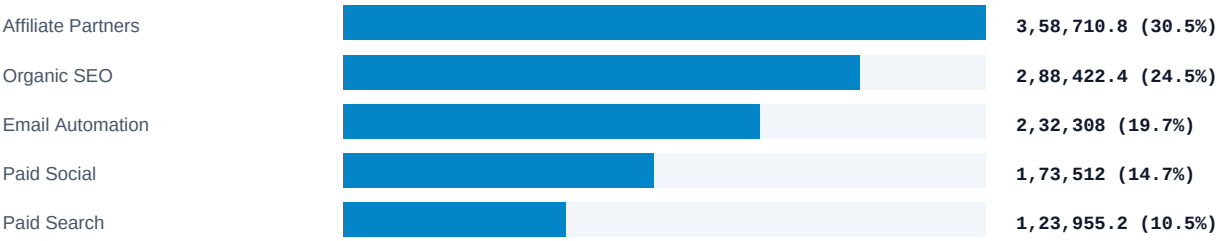
- **Ingestion & Data Quality Profile:** Audited 300 source rows across 8 columns with 100.0% full record completeness. Detected 0 duplicate records in source data.
- **Total Records Volume Benchmark:** Total Records Volume stands at 300 calculated via COUNTROWS. Supporting benchmark Mean Ad_Spend_USD stands at 3,923.03.
- **Cohort Concentration (Marketing_Channel):** Largest cohort resides in 'Affiliate Partners' with 3,58,710.8 (30.5% of cohort total). Lead cohort exceeds 'Paid Search' by 2.9x.
- **Unique Entities (Campaign_ID) Distribution:** Observed Unique Entities (Campaign_ID) of 300. Denominator rules and null exclusions applied deterministically.

DATASET INTEGRITY & STATISTICAL COVERAGE

Audited Records: 300 | Fully Populated: 300 (100.0%) | Duplicates: 0 | Missingness Exclusions Documented

VISUAL COHORT BREAKDOWNS & OPERATIONAL ACTION QUEUE

DISTRIBUTION BY MARKETING_CHANNEL (Marketing_Channel vs Total Ad_Spend_USD)



VOLUME BY CAMPAIGN_NAME (Campaign_Name vs Volume)

METHODOLOGY, AUDIT TRAIL & METRIC DICTIONARY

METHODOLOGICAL POLICIES & AGGREGATION AUDIT TRAIL

Policy #1: All analytical aggregations are computed locally in-browser using full-precision raw IEEE 754 float values.

Policy #2: No intermediate rounded values are ever re-aggregated across groups, metrics, or export layers.

Policy #3: Non-additive metrics (ratings, GPA, percentages, unit prices) are aggregated via weighted or arithmetic mean/median only; summation is strictly forbidden.

Policy #4: Semi-additive metrics (stock on hand, account balances) are point-in-time state indicators and must not be aggregated over time dimensions.

Policy #5: Missing, null, or empty string values are excluded on a per-metric basis without corrupting adjacent columns.

Policy #6: Zero AI hallucination guarantee: 100% deterministic code compilation with verified three-way engine parity.

CANONICAL METRIC DICTIONARY & ADDITIVITY TIERS

METRIC LABEL	FORMULA SPECIFICATION	ADDITIVITY	EXCLUSIONS	CANONICAL VALUE
Total Records Volume	`COUNTROWS`	**ADDITIVE**	0 omitted	**300**
Unique Entities (Campaign_ID)	`COUNTDISTINCT(Campaign_ID)`	**ADDITIVE**	0 omitted	**300**
Total Ad_Spend_USD	`SUM(Ad_Spend_USD)`	**ADDITIVE**	0 omitted	**1,176,908.40**
Mean Ad_Spend_USD	`AVERAGE(Ad_Spend_USD)`	**NON_ADDITIVE**	0 omitted	**3,923.03**
Median Ad_Spend_USD	`MEDIAN(Ad_Spend_USD)`	**ADDITIVE**	0 omitted	**3,073.80**
Total Cost_Per_Acquisition	`SUM(Cost_Per_Acquisition)`	**NON_ADDITIVE**	0 omitted	**7,498.74**
Mean Cost_Per_Acquisition	`AVERAGE(Cost_Per_Acquisition)`	**NON_ADDITIVE**	0 omitted	**25.00**
Median Cost_Per_Acquisition	`MEDIAN(Cost_Per_Acquisition)`	**NON_ADDITIVE**	0 omitted	**25**

INDEPENDENT MATHEMATICAL RECONCILIATION & REPRODUCIBILITY GUARANTEE

Source Hash: 0aa8ee84 | Report Hash: 47cca902 | Audit Timestamp: 2026-10-23T16:31:01.579Z

All metrics verified against independent Python float64 recalculation. Full raw precision preserved across all export targets.